A Traffic Anomaly Detection and Identification Approach Based on Multi-instance Learning

Dong Feng
Institute of Information Science, Beijing Jiaotong University, Beijing, China, dongfeng@bjtu.edu.cn, Corresponding author

Mangui Liang
Institute of Information Science, Beijing Jiaotong University, Beijing, China
mgliang@bjtu.edu.cn

Guangchao Wang
Qingdao New Generation Artificial Intelligence Technology Research Institute, Qingdao, China
wangguangchao@boyunvision.com

ABSTRACT

Traffic anomaly detection plays an important role in effective prevention and timely handling of traffic accidents. However, currently traffic anomaly detection is still in its infancy, mainly depending on manual intervention, which not only consumes a lot of manpower, but also is unfavorable in timeliness. According to the characteristics of urban road traffic scenes, this paper proposes a MFnet network structure based on multiple fiber modules, aiming to realize fast extraction and real-time calculation of video streaming features via group convolution and sparse connectivity. In addition, the weakly-supervised multi-instance learning method is introduced for the training on application of traffic anomaly detection model, which reduces the difficulty of labeling sample videos and improves the capacity of traffic anomaly detection in complex scenarios. Experimental results based on real traffic video data show that this method proposed herein, compared to the existing traffic anomaly detection methods, is good in terms of detection accuracy and recall rate, and is efficient in traffic anomaly detection in actual traffic scenarios.

CCS CONCEPTS

• **CCS –> Computing methodologies –> Modeling and simulation –> Model development and analysis –> Modeling methodologies**;

KEYWORDS

Video anomaly detection, machine learning, AI, feature extraction, multi-instance learning

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1 INTRODUCTION

Image processing, computer vision, deep learning and other computer technologies make it possible to break through the limitations in traditional video monitoring and realize intelligent analysis of video monitoring data, as well as active detection and real-time warning of traffic anomaly. They are of great significance to intelligent monitoring and analysis of urban traffic videos.

In urban traffic scenes, such behaviors as illegal parking, slow driving and converse driving are very likely to cause traffic accidents and congestion. Severe traffic accidents give rise to both personal injuries and huge economic losses. Timely detection of and quick and effective response to any traffic accident are vital to ensure urban traffic safety. Video-based traffic anomaly detection is always a key research direction of intelligent transportation. Traffic anomaly detection based on video monitoring is mainly composed of four basic steps, i.e., image preprocessing, basic event representation, construction of anomaly detection model, and anomaly judgment [13-15]. Basic event representation and construction of anomaly detection model play an important role in ensuring the efficiency and performance of traffic anomaly detection. Relying on the multi-fiber network structure based on semantic image segmentation, and the MFNet video identification algorithm, this paper achieves the extraction and classification of video image features while reducing the calculation amount of spatio-temporal deep neural networks. Also, the multi-instance learning method for the scoring network of traffic anomaly detection is introduced to iteratively optimize the classification results.

This paper mainly 1) expounds existing anomaly detection algorithms and their advantages and disadvantages. 2) Makes clear video fragmentation rules and designs the MFNet network structure to achieve rapid extraction of video image features. 3) Introduces the multi-instance learning method for the scoring network to maximize the positive and negative scores and improve the accuracy of traffic anomaly detection.

This paper mainly consists of five sections and summary. To be specific, Section 1 introduces the background and significance of traffic anomaly detection. Section 2 analyzes the existing research results, advantages and disadvantages of traffic anomaly detection. Section 3 expounds the design and application of MFNet network model. Section 4 elaborates on the application of weakly-supervised multi-instance learning method to the scoring network. Section 5 presents the experimental details and results. Finally, summary and outlook are given.

2 RELATED WORK

Compared with the classic target detection, video anomaly detection usually makes it impossible to collect sufficient and complete anomaly datasets. The conventional and effective supervised classification technique usually is not suitable for video anomaly detection (regarding the supervised classification technique, the data containing the information of both normal and anomalous events labeled are required for the purpose of training) [10-12]. By usage scenarios and conditions, existing video anomaly detection techniques fall into the following three categories:

- Semi-supervised video anomaly detection

Regarding semi-supervised video anomaly detection, the videos about normal events only can be used to construct a normal event model, and all events that deviate from this normal event model in the test stage will be identified as anomalous events. Semi-supervised video anomaly detection technique assumes that there is no prior information of anomaly, which most accords with the Open World Assumption for anomaly detection. Most of the existing solutions are based on such assumption. See related literature [1-5] for details of representative solutions.

- Maintaining the Integrity of the Specifications Unsupervised video anomaly detection

Any artificially labeled data (normal or anomalous events) are not used for unsupervised video anomaly detection, and the most differentiated anomaly is identified based on the properties and distribution characteristics of video event data. Compared to semi-supervised video anomaly detection, unsupervised video anomaly detection is closer to reality in scenario setting, but is much lower in the requirements for training data. In addition, unsupervised video anomaly detection as an emerging research direction in recent recent years is more difficult to be performed. See related literature [6-8] for details of representative work.

- Weakly-supervised video anomaly detection.

Regarding weakly-supervised video anomaly detection, few or weakly labeled videos containing anomalous event information may be used for training. With regard to weak labeling, anomaly pixel or video frame position is marked in a manner of imprecise labeling, and whether a video contains anomaly information is indicated at the coarse-grained level only (marking whether a video contains anomaly information, without need to mark the specific frame position of anomaly). Weakly-supervised video anomaly detection is closer to real life scenarios, which mainly overcomes the weakness of fully ignoring the prior information of known anomalous event in the mode of semi-supervised video anomaly detection. At present, few methods of weakly-supervised video anomaly detection as a new research direction are available. See related literature [9] for details of representative work.
This paper achieves a rapid extraction of image features of the video frame sequence via the MFNet network model, and introduces the multi-instance weakly-supervised learning method for traffic anomaly scoring, making an in-depth research on traffic anomaly detection.

3 THE DESIGN OF MFNET FOR FEATURE EXTRACTION

The 3D convolutional neural network (CNN) is preferred for extraction and classification of video image features. Different from the 2D CNN, the 3D CNN uses the 3D convolution kernel across time and space for the purpose of network convolution. Although the 3D CNN can extract image features very accurately, it brings a huge overhead of convolution operation, and also an exponential increase in input/output tensor dimension in contrast with the 2D CNN. In order to reduce the computation burden of the spatio-temporal deep neural networks and ensure the inference speed approximate to that of the 2D CNN while keeping the classification accuracy, this paper proposes the concept of Multi-Fiber network, divides the complex network into multiple lightweight networks via the sparse connection structure, and achieves information fusion between different fibers through the multiplexer module.

This section first introduces the multi-fiber unit, the basic unit of 3D CNN model. The multi-fiber unit minimizes the internal connection layers of network and improves the model efficiency, having good compatibility and adaptability to 2D CNN and 3D CNN. The multi-fiber unit is embedded into the 2D CNN to illustrate its effectiveness and evaluate its accuracy in image identification. This section applies the multi-fiber unit to the 3D CNN, discussing the specific solution.

The multi-fiber unit is designed based on optimization of the highly modular residual unit, which facilitates the model training and deployment. The traditional residual unit learns features via two convolutional layers, which is clear and easy to understand, but brings a huge computation burden, as shown in Figure 1(a). The number of convolutional connections of the multi-fiber unit is expressed by the following formula. Where, M_{in} denotes the number of input channels, M_{mid} represents the number of channels in the intermediate layer of the residual unit, and M_{out} is the number of output channels:

$$Connections = M_{in} * M_{mid} + M_{mid} * M_{out} \tag{1}$$

The constants arising from the changes in connection relationship are taken into account only, while such constants as input feature dimension and convolution kernel dimension are ignored. Thus, the aforesaid formula is simplified. The formula (1) shows that the number of connections is proportional to the square of network width. In other words, computational overhead will increase by k^2if the network width of structural unit increases by k times. In order to reduce network connections and overall computational overhead, the complex residual unit is sliced and converted into N separable parallel channels, which is the so-called modular fibrosis. After fibrosis, each channel in the residual unit becomes independent which is shown in Figure 1. Network width of structural unit remains unchanged, but the number of connections is reduced to 1/N of the original, as shown in formula (2):

$$\begin{aligned} Conns &= N * (M_{in}/N * M_{mid}/N + M_{mid}/N * M_{out}/N) \\ &= (M_{in} * M_{mid} + M_{mid} * M_{out})/N \end{aligned} \tag{2}$$

In order to realize the fusion and sharing of feature information between different channels, the multiplexer module is accessed prior to convolution operation, which amplifies and redirects the

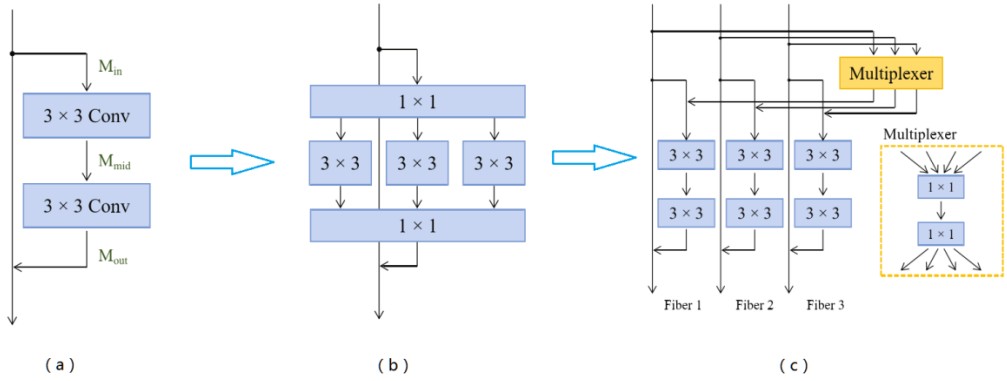


Figure 1: Process of change from ResNet network to multi-fiber network

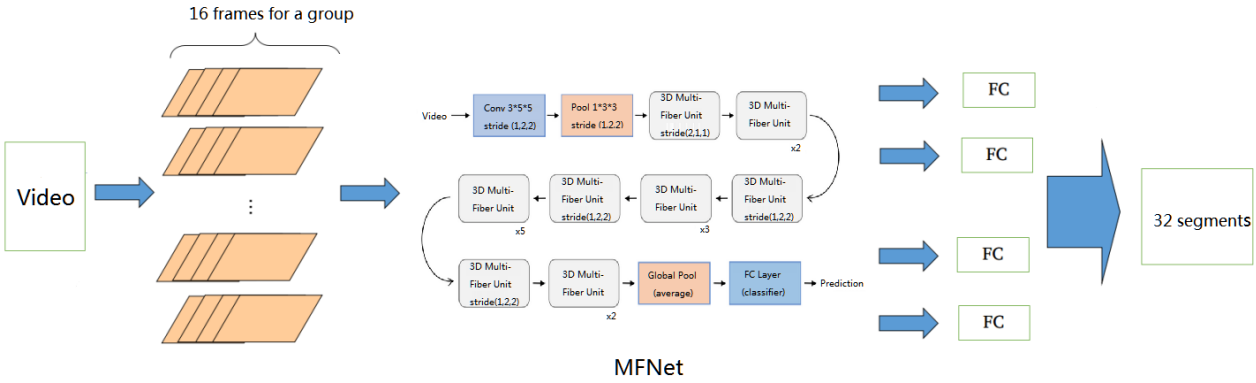


Figure 2: Multi-fiber network structure based on 3D convilution

multi-channel feature information, playing the role of routing distribution. The multiplexer module first performs channel dimension reduction through the 1*1 convolution kernel, so as to realize the information interaction between different channels and make up for the information loss generated from slicing. Next, the number of input channels is restored through the dimension upgrade of 1*1 convolution kernel.

The 3D CNN based on MFNet is designed to extract video features and detect anomalies. MFNet is of ResNet-50 network structure, and the residual unit is replaced with the multi-fiber unit proposed herein. After the features of each video are extracted by 16 frames, all the feature vectors of a complete video are integrated into a 32×768 feature vector and then sent to the MFNet network for feature extraction. If the video has less than 32 feature vectors, adjacent feature vectors will be used for filling; otherwise, 32 feature vectors will be screened out in the way of uniform sampling. Then the feature vectors are inputted to the FC layer for anomaly scoring, as shown in Figure 2. The multi-fiber network structure proposed herein can reduce the 3D CNN channel connections and computation overhead, and also improve the efficiency of 2D CNN model. In order to prove the general applicability of the multi-fiber network structure, the experimental part of this paper expounds the improvement of 2D image classification in performance and accuracy.

4 WEAKLY-SUPERVISED MULTIPLE INSTANCE LEARNING

After extraction of video image features via MFNet, 32 segments of fully connected features are sent to the scoring network for classification and scoring of anomaly segments. Unsupervised learning method is usually adopted for traditional traffic anomaly detection, which can reduce the difficulty of data labeling. However, it is almost impossible to collect all normal events and anomalous events in reality, and the judgments on the same event may vary in different scenarios. Therefore, accurate classification results are unavailable after training and scoring via the unsupervised learning method. This paper adopts the weakly-supervised multi-instance learning method for anomaly classification and scoring. Positive and negative bags are defined in Figure 3. Each training video is divided into 32 segments evenly. If all the 32 video segments are normal events, the video constitutes a negative bag. If one or more anomalous events exist in 32 video segments, the video is defined as a positive bag. Compared to the negative bag, the positive bag needs no additional time/space labeling information to identify the specific location of anomaly.

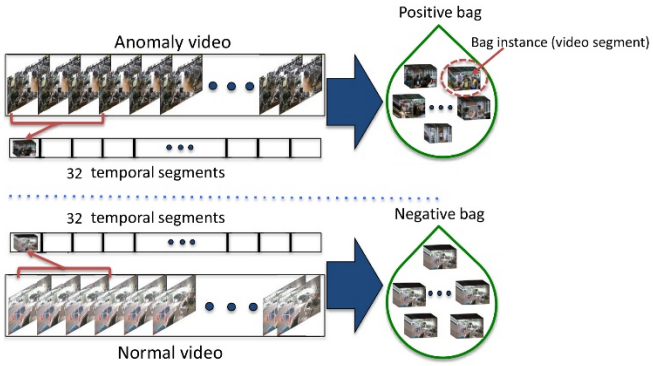


Figure 3: Division of weakly-supervised positive and negative bags

The scoring network is based on the 3-layer fully connected neural network as shown in Figure 4, via which each video segment is scored. In order to avoid over-fitting and improve the generalization capacity of the scoring network, the dropout layer is added to the scoring network, and some feature detectors are randomly stopped. The video feature segments of positive and negative bags are inputted into the network model, and each feature segment is scored through the scoring network. The feature segment with the highest score is selected from the positive bag as the potential anomaly instance, and the feature segment with the highest score is selected from the negative bag as the non-anomaly instance. The model parameters are trained based on the scores of the two instances. The hinge-loss is used as the loss function to maximize the score gap between the positive instance and the negative instance. The formula is as follows:

$$l(\beta_a, \beta_n) = \max(0, 1 - \max_{i \in \beta_a} f(v_a^i) + \max_{i \in \beta_n} f(v_n^i)) \tag{3}$$

Where, f() represents the scoring output. When the scores of anomalous event and normal event are 1 and 0, the loss value reaches the minimum 0, otherwise it is greater than 0. Therefore, the loss function can maximize the distance space between positive and negative instances. In order to improve the accuracy of anomaly detection, two constraints are added to the loss function, namely smoothness constraint and sparsity constraint. First, all feature segments of each positive bag or negative bag come from the same video, so the scores of feature segments should change smoothly. The square of the difference between two adjacent feature segments is minimized to ensure the smooth transition of feature values. Second, only few anomaly instances usually exist in a positive bag, so smoothness constraint may make all instances in a positive bag tend to get higher scores. Therefore, the loss is added to separately constrain the overall score. The overall expression is shown in formula 4:

$$l(\beta_a, \beta_n) = \max(0, 1 - \max_{i \in \beta_a} f(v_a^i) + \max_{i \in \beta_n} f(v_n^i))$$

$$+ \lambda_1 \sum_{i}^{(n-1)} (f(v_a^i) - f(v_a^{i+1}))^2 + \lambda_2 \sum_{i}^{n} f(v_a^i) \tag{4}$$

Finally, regularization items are added to obtain the objective function expression as follows:

$$L(W) = l(\beta_a, \beta_n) + \lambda_3 \|W\|_F \tag{5}$$

5 EXPERIMENTAL RESULTS

5.1 Implementation Details

First, performance indicators of the MFNet network model and the recent advanced algorithm model were evaluated, with Kinetics, UCF-101 and HMDB51 as public datasets. Then, the scoring network based on multi-instance learning was used to detect and evaluate anomalous events. All experiments are based on the pytorch framework. Before the video stream entered the network model, the input pixel size was uniformly adjusted as 224*224, and the frame rate was set to 30fps. Then, video segments were generated by every 16 frames and inputted into the MFNet network. At the network output terminal, 16-frame image features were integrated into a 768-dimensional feature vector. Then, 32 feature segments obtained after video sampling were inputted to the neural network composed of three FC layers. The three FC layers consist of 512, 32 and 1 feature detection units, respectively. The dropout layer was added between the FC layers, and some feature information was randomly discarded to ensure the generalization capacity of the model. Regarding the parameters of loss function, the coefficient of correspondence between smoothness constraint and sparsity constraint was set as$\lambda_1 = \lambda_2 = 6 * 10^{-5}$, and the regularization coefficient was set as $\lambda_3 = 0.02$to output the best performance index.

5.2 Results and evaluation

First, effectiveness of the MFNet model on video feature learning was verified, and the large-scale data set Kinetics was used to evaluate the model (Kinetics data set consists of 300,000 video segments and contains 400 categories of action). The initial learning rate of the model was set to 0.1, which declined by 0.0001 gradually during the training process. The stochastic gradient descent (SGD) was adopted to optimize each batch of training. The cluster composed of 64 GPUs was used for model training. Figure 5 shows the curve of training and verification accuracy. Seen from this figure, the network model could quickly converge, and only about 36,000 iterations were needed to complete the entire training.

Table 1 shows the model video identification results generated from different algorithm trainings under the conditions of large-scale data set Kinetics. According to the experimental data shown in the following table, the MFNet algorithm, in contrast with other algorithms, significantly improves in the number of weight parameters, the number of floating-point calculations, and the accuracy of Top-1. Few multi-fiber bundles are needed for the MFNet algorithm, which can greatly reduce the computational overhead while ensuring the identification accuracy. The use of multi-fiber network structure reduces the overall computational overhead to 11.1G only, equaling to the computational overhead of ordinary 2D-CNN. MFNet enables the highest identification accuracy of Top-1/Top-5. In terms of size, MFNetR algorithm is only 1/8 of R(2+1)D-RGB algorithm, but is 13 times faster than the R(2+1)D-RGB algorithm

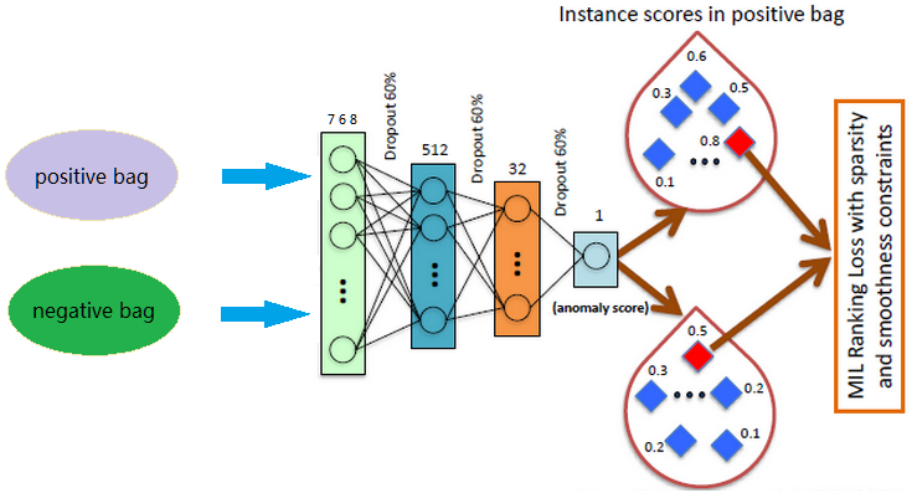


Figure 4: Flow chart of weakly-supervised anomaly detection

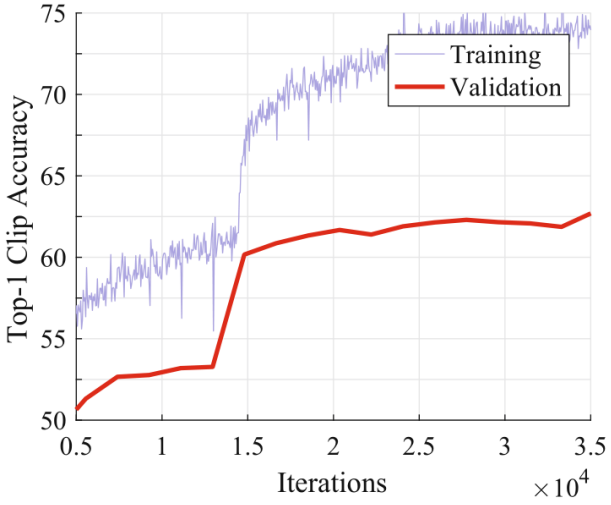


Figure 5: Identification accuracy of MFNet under the conditions of Kinetics dataset

in computation and inference speed, with the improvement of Top-1 accuracy by 0.8%. In addition, the MFNet model of multi-fiber network structure minimizes the use of GPU memory resources in the training or inference process.

Through the weakly-supervised multi-instance learning method, normal events and anomalous events can be identified in the constant iteration process. Through this method, the network can automatically learn a large number of labeled videos with positive and negative instances, and then predict the locations of anomaly segments in the video. The network can output high scores for anomaly video segments in the iterative training process. Figure 6 illustrates the changes of anomaly scores with an increase in iteration during the training of videos containing anomalous event information. When the number of iterations reaches 1,000, the network gives high scores to anomalous and normal segments.

Table 1: Performance comparison between MFNet algorithm and several classic algorithms under the conditions of Kinetics dataset

Method	Params	FLOPs	Top-1	Top-5
Two-Stream	12.5M	-	62.2%	-
ConvNet+LSTM	9M	-	63.3%	-
S3D	8.8M	66.4G	69.4%	89.1%
I3D-RGB	12.1M	107.9G	71.1%	89.3%
R(2+1)D-RGB	63.6M	152.4G	72.0%	90.0%
MF-Net(Ours)	7.8M	10.6G	73.8%	91.4%

After iterations for 3,000 times, the network gradually separates the anomalous and normal segments, anomaly segments still maintain high scores, and the scores of normal segments begin to decline. However, an improvement is still needed in duration and separation of anomaly segments, and some normal segments still exist in the anomaly segment score interval. When the numbers of iterations reach 5,000 and 8,000, the network can locate anomaly segments in the video more accurately after learning more video instances. Although the information in any video segment is not used, the location of an anomaly segment in the video can still be predicted through the MFNet algorithm according to the anomaly score.

Finally, the MFNet algorithm was analyzed in actual traffic scenes. First, an video segment of a typical traffic accident was selected, as shown in Figure 7. Traffic anomaly is shown between the 9th frame and the 12th frame of the video. Based on the MFNet algorithm, traffic anomaly was detected in a timely manner. As shown in the detection curve, a very high anomaly score is given in case of any traffic anomaly. The curve is relatively flat from the first frame to the ninth frame, where the anomaly scores are small and within the given thresholds. However, the detection curve falls from the 13th frame to the 15th frame. The MFNet algorithm is effective to detect any traffic anomaly in a timely manner, but the problem of continuous anomaly tracking still needs to be

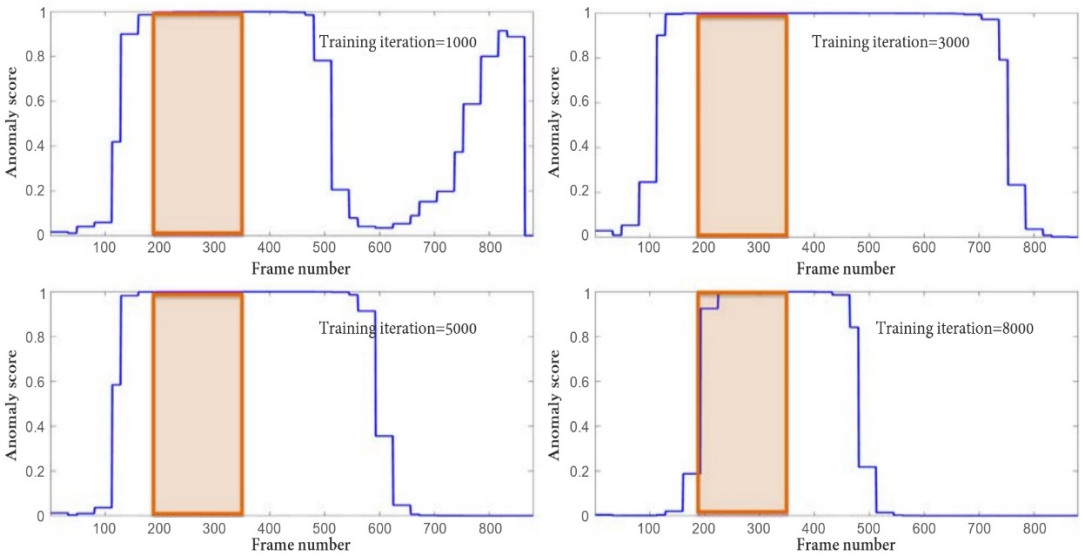


Figure 6: Changes of training video anomaly scores with an increase in iterations



Figure 7: Identification of anomaly in the video through the weakly-supervised method

solved. Long Short-term Memory Networks like LSTM and MFNet network need to be comprehensively integrated and analyzed to ensure complete monitoring of anomalous events.

6 CONCLUSIONS

A deep learning method is hereby proposed to achieve traffic anomaly detection. The complexity and diversity of anomaly make it difficult to achieve anomaly detection only by relying on normal event data. Therefore, two types of data are used for model training. In order to avoid massive labeling work related to the training video data, multi-instance learning network structure is adopted for instance scoring. Little labeling work is needed, but the accuracy of event classification and scoring is not affected. The experimental results prove that the new traffic anomaly detection method is significantly better than traditional detection algorithms. The future research will focus on the association of various traffic conditions and parameters with the occurrence of anomalous events, so as to realize better early warning for any possible traffic anomaly.

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